

Debug Your Application with Strace

When a program seems to hang, errors out, or takes too long to respond, have you wondered what's happening? Or tried to find out what is going on underneath, where it is stuck or why it takes so long to complete the task? Strace answers all these questions.



Strace allows you to trace the execution of a program and its interaction with the operating system. It does this by monitoring and displaying information about the signals and system calls a binary makes—you don't need the source code, or debug symbols.

A system call is the gateway or API for a user space program to talk to the operating system, i.e., the kernel space. For example, calls such as `open()`, `socket()`, `wait()`, `clone()`, `write()`, etc. are system calls.

Let us start with an example:

```
$ strace ping google.com
execve("/bin/ping", ["ping", "google.com"], [/* 44 vars */])
= 0
brk(0)                                = 0x7fb487663000
....
```

The first line shows the program being executed along with the environment variables. This is actually the `execve` system call. To know more about this call, run `man execve`. Man pages are available for all the system calls that we will encounter and are a very good resource to learn about them.

```
access("/etc/suid-debug", F_OK)      = -1 ENOENT (No such
file or directory)
```

```
map(NULL, 4096, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_
ANONYMOUS, -1, 0) = 0x7f3b6cfa000
access("/etc/ld.so.preload", R_OK) = -1 ENOENT (No such
file or directory)
open("/etc/ld.so.cache", O_RDONLY) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=92520,...}) = 0
map(NULL, 92520, PROT_READ, MAP_PRIVATE, 3, 0) =
0x7f3b6cfa3000
close(3) = 0
```

Next, we see that it is trying to load the file `/etc/ad.so`. Use the `man` command to learn more about each call. The value after the equal sign is the return code. For example `-1 EONENT` is returned for the `access` call, because the file does not exist. For the `open` call, we see a single number 3 returned. This is the file descriptor that would be used in subsequent calls to refer to that file (like in the `mmap` and `close` commands).

```
open("/lib64/libc.so.6", O_RDONLY) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\260\3
55a\374\0\0\0..., 832) = 832
fstat(3, {st_mode=IFREG|0755, st_size=1912928, ...}) = 0
fcntl(NULL, F37768, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_
```